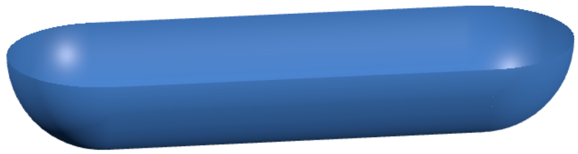
B4 past final worksheet

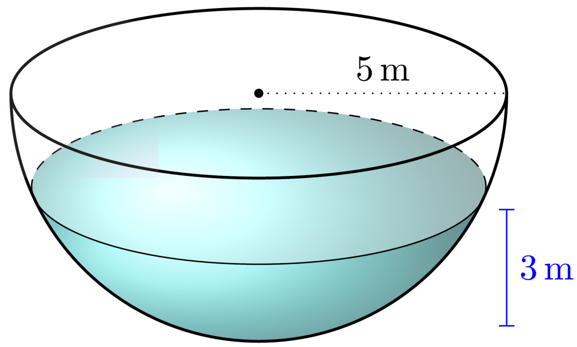
1. (Fall 2024, #5) We intend to lift a basket to the top of a building by means of a rope. Initially, half of the rope is lying on the ground and the other half is hanging from the top. The basket is attached to the end of the rope. The total length of the rope is 50m, the rope's linear density is 1 kg/m, and the mass of the basket is 2 kg. Find the amount of work needed to lift the basket to the top of the building.

2. (Spring 2024, #5) A container full of fuel is in the shape of the region in the first quadrant bounded by $y = (x+1)^{1/2}$, $x = 0$ and $y = 4$ rotated about the y-axis. Find a definite integral whose value is the work needed to pump the fuel out of the tank. Assume the values of x and y are in meters, the density of the fuel is ρ kg/m^3, and the acceleration due to gravity is g m/s^2. You do not need to evaluate the integral.

3. (Fall 2023, #5) A tub filled with water has the shape of a half-cylinder with hemispherical ends, as pictured below. The tub is 5m tall and the base has a length of 20m.Set up an integral expressing the work required to empty the tank by pumping all of the water out of the top. You may assume the density of water is $\rho = 1000$ kg/m^3 and the acceleration due to gravity is $g = 10$ m/sec^2. **You do not need to evaluate the integral.**



4. (Spring 2023, #3) A tank has the shape of a hemisphere with a radius of 5 meters. It is filled with water to a height of 3 meters. Set up, **but do not evaluate**, an integral that gives the work required to empty the tank by pumping all of the water to the top of the tank. (The density of water is 1000 kg/m^3 and the gravitational constant is g).



5. (Fall 2022, #4) A tub with a circular bottom is created by rotating the curve $y = \ln x$ about the y-axis between $x = 1$ and $x = e^2$.

(a) Write down an expression for the **surface area** of the tub. You do **not need to evaluate** the integral.

(b) The tub is filled with water. Find the **work** required to pump the water out from the top, assuming x and y are measured in meters. You may use that the density of water is $\rho = 1000$ kg/m^3, and $g = 10$ m/s^2.